ASSIGNMENT I

ONE-PAGE PROPOSAL

Smart Piezoelectric Floors

Harvesting usable electricity from everyday footsteps

01 EXISTING TECHNOLOGY

Piezoelectric technology uses materials that produce an electrical charge when pressure or mechanical force is applied. In a piezoelectric floor, the force of a footstep is converted into electricity. Ordinary walking can therefore become a source of power.

02 PROBLEM / LIMITATION

The energy from each step is small, and the floors are expensive to install and maintain. Output is often too limited to run larger devices or to supply a building. Without more efficient materials and a way to store what is generated, much of the energy from foot traffic is lost.

03 TARGET USERS

Universities, shopping malls, train stations, airports, and other crowded public spaces are the main users. These places receive thousands of footsteps every day, but current systems do not give them enough reliable power. Facility managers are directly affected by the low output and by the cost of installation and maintenance.

04 PROPOSED ENHANCEMENT

The proposed enhancement is a smart piezoelectric floor that combines more efficient piezoelectric materials with energy storage, such as rechargeable batteries or supercapacitors. Smart sensors would monitor production and identify the areas with the highest foot traffic, so the floor can be placed where it is most useful. Stored electricity would power LED lights, digital information displays, and small sensors, putting everyday human movement to practical use.

05 EXPECTED IMPACT

Busy public buildings could supply part of the electricity used by lights, displays, and sensors from people walking on site. That would lower grid consumption, reduce operating costs, and make these facilities less dependent on conventional power. Foot-traffic data would also help managers improve lighting, information, and maintenance in the busiest areas, which benefits both operators and visitors.

06 SDG CONNECTION

The enhancement supports SDG 7, Affordable and Clean Energy, by turning human movement into usable electricity. It supports SDG 11, Sustainable Cities and Communities, by making high-traffic public infrastructure more energy efficient. It also supports SDG 13, Climate Action, by lowering demand for conventional electricity in lighting and information systems.

7 Affordable and Clean Energy

11 Sustainable Cities and Communities

13 Climate Action